Sums of Vector Spaces.

Def. (14.1) Let V be a vect. space over K and $\mathcal{U}, W \subseteq V$ linear subspaces. We define the sum $\mathcal{U}+W$ of $\mathcal{U}, W$ as $\mathcal{U}+W := \{u+w \mid u \in \mathcal{U}, w \in W\}$.

In a similar way we can define sums of more than two subspaces: if $\mathcal{U}_1, \dots, \mathcal{U}_r \subseteq V$ are subspaces define $\mathcal{U}_1 + \dots + \mathcal{U}_r := \{u_1 + \dots + u_r \mid u_1 \in \mathcal{U}_1, \dots, u_r \in \mathcal{U}_r\}$.

Prop. (14.2) Let $\mathcal{U}, W \subseteq V$ subspaces. Then:

1) $\mathcal{U}+W = Sp(\mathcal{U} \cup W)$. In particular $\mathcal{U}+W \subseteq V$ is a subspace and $\mathcal{U}, W \subseteq \mathcal{U}+W$ are subspaces of $\mathcal{U}+W$.

2) If $\mathcal{U}, W$ are f. dim. then so is $\mathcal{U}+W$. Moreover one can find a basis for $\mathcal{U}+W$ as follows.

   (a) Choose a basis $p_1, \dots, p_k$ for $\mathcal{U} \cap W$, where $k = \dim(\mathcal{U} \cap W)$.

   (b) Extend $p_1, \dots, p_k$ to a basis $p_1, \dots, p_k, u_1, \dots, u_{\ell-k}$ of $\mathcal{U}$, where $\ell = \dim \mathcal{U}$.

   (c) Exten $p_1, \dots, p_k$ to a basis $p_1, \dots, p_k, w_1, \dots, w_{m-k}$ of W, where $m = \dim W$.

   Then $p_1, \dots, p_k, u_1, \dots, u_{\ell-k}, w_1, \dots, w_{m-k}$ is a basis for $\mathcal{U}+W$.

3) In particular, we have $\dim(\mathcal{U}+W) = \dim \mathcal{U} + \dim W - \dim(\mathcal{U} \cap W)$.

Proof. 1) clearly $\mathcal{U}+W \subseteq Sp(\mathcal{U}\cup W)$, by the characterization of $Sp(\mathcal{U}\cup W)$ as the set of all linear combin. of elements from $\mathcal{U}\cup W$. But we also have $Sp(\mathcal{U}\cup W) \subseteq \mathcal{U}+W$.

Indeed, if $v \in Sp(\mathcal{U}\cup W)$, then $v = \underbrace{\sum_{i=1}^{s} a_i u_i}_{\in \mathcal{U}} + \underbrace{\sum_{j=1}^{r} b_j w_j}_{\in W}$, with $a_i, b_j \in K$, $u_i \in \mathcal{U}$, $w_j \in W$.

$\Rightarrow v \in \mathcal{U}+W$. This proves $Sp(\mathcal{U}\cup W) \subseteq \mathcal{U}+W$.

Putting all together we have $\mathcal{U}+W = Sp(\mathcal{U}\cup W)$.

The rest of the statements in 1 follow immediately.

2) We first claim that $p_1,\dots,p_k, u_1,\dots,u_{l-k}, w_1,\dots,w_{m-k}$ are lin. indep. Indeed, if

$$a_1p_1+\dots+a_kp_k+b_1u_1+\dots+b_{l-k}u_{l-k}+c_1w_1+\dots+c_{m-k}w_{m-k}=0.$$

Write $v := \underbrace{c_1w_1+\dots+c_{m-k}w_{m-k}}_{\in W} = \underbrace{-(a_1p_1+\dots+a_kp_k+b_1u_1+\dots+b_{l-k}u_{l-k})}_{\in \mathcal{U}}$.

$\Rightarrow v \in \mathcal{U}\cap W$.

$\Rightarrow$ We can also write $v = \alpha_1p_1+\dots+\alpha_kp_k$ for some $\alpha_1,\dots,\alpha_k \in K$.

But $v = c_1w_1+\dots+c_{m-k}w_{m-k}$. And $p_1,\dots,p_k,w_1,\dots,w_{m-k}$ are a basis for W. $\Rightarrow \alpha_1=\dots=\alpha_k=0$, $c_1=\dots=c_{m-k}=0$ and $v=0$. But now $0=v=-(a_1p_1+\dots+a_kp_k+b_1u_1+\dots+b_{l-k}u_{l-k})$, and $p_1,\dots,p_k,u_1,\dots,u_{l-k}$ is a basis for $\mathcal{U}$.

$\Rightarrow a_1=\dots=a_k=0$, $b_1=\dots=b_{l-k}=0$. This concludes the proof that $p_1,\dots,p_k,u_1,\dots,u_{l-k},w_1,\dots,w_{m-k}$ are lin. independent.

Now, clearly $p_1, \dots, p_k, u_1, \dots, u_{l-k}, w_1, \dots, w_{m-k}$ span $U+W$ (exc. check this!). $\Rightarrow$ $p_1, \dots, p_k, u_1, \dots, u_{l-k}, w_1, \dots, w_{m-k}$ is a basis for $U+W$.

3) $\dim(U+W) = k+(l-k)+(m-k) = l+m-k = \dim U + \dim W - \dim(U \cap W)$.

$\blacksquare$

Corollary (14.3) Let $U, W \subseteq V$ be two f. dim. linear subspaces of a vect. space V. The following are equivalent:

1) $\dim(U+W) = \dim U + \dim W$.
2) $\dim(U \cap W) = 0$.
3) $U \cap W = \{0\}$.
4) Every $v \in U+W$ can be written in a <u>unique</u> way as $v = u+w$ with $u \in U$, $w \in W$.
5) If $u+w=0$, where $u \in U$, $w \in W$ then $u=w=0$.

Proof. "1$\Leftrightarrow$2" follows from the previous Proposition (14.2).

"2$\Leftrightarrow$3" because there is only one lin. subspace of dim 0, the space $\{0\}$.

"3 $\Rightarrow$ 4": By Def. 14.1, every $v \in U+W$ can be written as $v = u+w$. To see the uniqueness, suppose that $v = u+w = u'+w'$, where $u, u' \in U$, $w, w' \in W$.

$\Rightarrow$ $u-u' = w'-w \in U \cap W = \{0\}$. $\Rightarrow$ $u' = u$ and $w' = w$.

"4 $\Rightarrow$ 5": $0 = 0+0$, so if $0 = u+w$ and 4 holds, then $u=0, w=0$.

"5 ⇒ 3": Let $v \in \mathcal{U} \cap W$. We have $0 = \underset{\mathcal{U}}{\underset{\cap}{v}} + \underset{W}{\underset{\cap}{(-v)}}$.

If 5 holds, we have $v = 0$. ∎

Def. (14.4) Let V be a vect. space over K and $\mathcal{U} \subseteq V$ a subspace. A subspace $W \subseteq V$ is called a complement of $\mathcal{U}$ if $\mathcal{U} + W = V$ and $\mathcal{U} \cap W = \{0\}$. (Note that $\mathcal{U}, W$ satisfy all the five equivalent statements from the corollary).

Prop. (14.5) Let V be a f.dim. vect. space and $\mathcal{U} \subseteq V$ a subspace. Then $\exists$ a subspace $W \subseteq V$ which is a complement to $\mathcal{U}$.

Proof. Choose a basis $u_1, \dots, u_\ell$ to $\mathcal{U}$, where $\ell = \dim \mathcal{U}$. Extend this basis to a basis of V: $u_1, \dots, u_\ell, w_1, \dots, w_m$, where $\ell + m = \dim V$. Take $W := Sp(w_1, \dots, w_m)$.

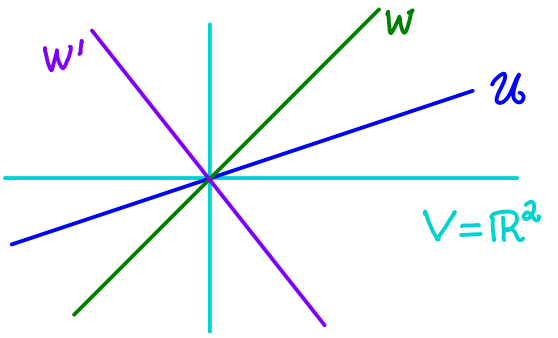
Clearly $\mathcal{U} + W = V$. And $\mathcal{U} \cap W = \{0\}$, b.c. if

$$\sum_{i=1}^{\ell} a_i u_i = \sum_{j=1}^{m} b_j w_j \Rightarrow \sum_{i=1}^{\ell} a_i u_i + \sum_{j=1}^{m} (-b_j) w_j = 0.$$ But

$u_1, \dots, u_\ell, w_1, \dots, w_m$ are lin. indep. $\Rightarrow a_1 = \dots = a_\ell = 0 = b_1 = \dots = b_m$.

$\Rightarrow v = 0$. ∎

Important remark. If $\mathcal{U} \subsetneq V$, then there many different subspaces $W \subseteq V$ which are complements of $\mathcal{U}$. In general there is no canonical choice for a complement W.



Both W and W' are complements of $\mathcal{U}$.